Neural Operators for Cardiac Field Reconstruction from Simulated Magnetic Sensors

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Abstract

Learning physical state from remote sensors requires a model to invert a measurement process that can attenuate and spatially mix the underlying signal. We study this problem for cardiac magnetic sensing using 500 independent openCARP simulations on a controlled ventricular tissue sheet. A Biot–Savart forward calculation converts simulated electrophysiology into 301-frame magnetic sensor sequences, and Fourier neural operator (FNO) and DeepONet models reconstruct local activation time (LAT) and action potential duration (APD). On 75 grouped test simulations, FNO attains LAT and APD90 R^2 of 0.954 and 0.960, while DeepONet attains 0.968 and 0.970. With the trained checkpoints frozen, we recompute both magnetic channels at tissue distances of 35, 50, and 65 mm. DeepONet remains within 0.0013 of its nominal LAT and APD90 scores at every tested distance, whereas FNO declines at both alternative distances, losing up to 0.0737 LAT R^2 and 0.1082 APD90 R^2 at 65 mm. Additional region evaluation shows that accurate smooth fields do not guarantee reliable recovery of compact conduction abnormalities. The study provides a reproducible test bed for physical sensor inversion and identifies acquisition geometry and task-specific evaluation as distinct requirements for learned field reconstruction.

1 Introduction

Physical sensors rarely observe the state of interest directly. Their measurements are transformed by geometry, material properties, distance, noise, and the governing physics of signal propagation. An effective learning system must therefore recover a spatial field from a sensor field while remaining stable under plausible changes to the acquisition arrangement. This structure appears in robotics, geophysics, medical imaging, and other inverse problems. It is especially clear in magnetocardiography, where sensors outside the body measure weak magnetic fields generated by cardiac electrical activity [1–4].

The magnetic inverse problem is difficult because distinct internal sources can produce similar external measurements, and because distance suppresses fine spatial detail [5]. Related work in electrocardiographic imaging has demonstrated that noninvasive measurements can recover useful activation patterns in simulation and in patients [6–9]. These successes also show why evaluation must match the intended scientific quantity. A field can agree globally while a small region or boundary is misplaced.

Neural operators provide a natural representation for this setting. They learn mappings between functions rather than mappings between fixed vectors [10]. The Fourier neural operator (FNO) represents global integral kernels with spectral convolutions [11]. DeepONet separates an encoding of the observed function from a coordinate-dependent query network [12]. Both can map a time-

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dependent sensor field to a spatial electrophysiology field, but their different treatment of input geometry may produce different behavior when the physical acquisition changes.

We build a controlled benchmark around three linked components: a cardiac tissue simulator, an explicit magnetic measurement calculation, and learned inverse operators. The benchmark contains 500 independent electrophysiology cases drawn deterministically from a 5,000-case generated corpus. Each case produces 301 magnetic frames on a 32×32 sensor grid and target maps of local activation time (LAT), APD80, APD90, and abnormal tissue. Simulation identifiers define the split, so measurement rows from one simulated case cannot cross partitions.

Our main finding is positive. Both FNO and DeepONet reconstruct activation and repolarization fields accurately on 75 unseen simulations, with DeepONet reaching pooled R^2 near 0.97 for both endpoints. We then freeze the same field checkpoints and change sensor-to-tissue distance. DeepONet changes by less than 0.0013 across 35, 50, and 65 mm, while FNO is sensitive in both directions away from its nominal training distance. Because normalization is performed within each example and geometry remains planar, this is a controlled sensitivity result rather than evidence of clinical or anatomical generalization.

The paper makes four contributions. First, it defines a reproducible physical-sensor inversion benchmark linking openCARP electrophysiology to direct magnetic field formation. Second, it compares two operator families under grouped model selection and held-out evaluation. Third, it tests frozen checkpoints after a physical change to acquisition geometry. Fourth, it evaluates compact abnormal-region recovery separately from global reconstruction, revealing that the two objectives can diverge. Code and verification scripts are available in the GitHub code repository, while selected checkpoints, test arrays, bootstrap outputs, and provenance records are available in the OSF results archive.

2 Related work

Cardiac field sensing and inverse reconstruction. Magnetocardiography was introduced as a contactless measurement of fields generated by cardiac activity [1, 2]. Subsequent work examined its clinical potential and the computational challenges of source inference [3, 5, 4]. Electrocardiographic imaging provides a more mature example of recovering cardiac activation from remote measurements [6–8]. Comparative studies now include both physics-based and data-driven reconstruction [9]. Our study is narrower than these clinical systems. It uses synthetic relative-unit magnetic observations on a planar sheet to isolate the behavior of learned inverse operators.

Biophysical simulation. Cardiac electrophysiology models enable controlled intervention on conductivity and ionic parameters [13]. The Bueno-Orovio–Cherry–Fenton model gives a compact ventricular action potential description suitable for tissue simulation [14]. openCARP supplies an open environment for electrophysiology simulation [15]. Verification studies emphasize the importance of solver, mesh, and model choices when interpreting simulated cardiac behavior [16–18]. We use simulation as a controlled generator of paired source and sensor fields. We do not treat it as a substitute for validation with measured human or animal data.

Learning operators for physical systems. Neural operators are designed to approximate maps between discretized functions and have shown strong performance on partial differential equation families [10]. FNO uses truncated Fourier modes to implement global mixing efficiently [11]. DeepONet uses a branch network for the input function and a trunk network for output coordinates, providing an explicit query interface [12]. Physics-informed neural networks and physics-informed DeepONets add residual constraints when equations and differentiable operators are available [19, 20]. Our models are supervised. The cardiac equations and magnetic forward model generate the data, but are not enforced in the loss.

Evaluation of localized physical state. Aggregate scores can hide errors on small structures. Dice overlap is a standard region metric [21], but its interpretation depends on target size and task [22]. Biomedical challenges have shown that metric choice can change rankings and scientific conclusions [23, 24]. Similar concerns motivate explicit reporting of leakage controls, independent sampling units, and uncertainty in scientific machine learning [25–27]. Our evaluation therefore distinguishes field reconstruction, region overlap, condition-specific behavior, and sensitivity to sensor distance.

3 Physical sensor benchmark

3.1 Electrophysiology and controlled tissue perturbations

The upstream corpus contains 5,000 successful cardiac tissue simulations. We selected a deterministic, condition-stratified pilot of 500 independent cases: 75 homogeneous, 125 with altered repolarization, 125 with slowed conduction, and 175 with both changes. Each uses a fixed 10 by 10 mm ventricular sheet and the Bueno-Orovio–Cherry–Fenton model in openCARP's monodomain environment [14, 15]. Stimulus edge, perturbation location and size, and physiological parameters vary. Nineteen prepacing beats precede the saved final beat at a 1,000 ms cycle length. Appendix C gives mesh and parameter details. Labels describe simulator interventions, not clinical diagnoses.

The simulator produces transmembrane voltage $V_m(\mathbf{x}, t)$ on a triangular tissue mesh. The magnetic generator uses unit isotropic scaling to calculate an equivalent-current proxy

$$\mathbf{J}_{\mathrm{eq}}(\mathbf{x}, t) = -\nabla V_m(\mathbf{x}, t). \tag{1}$$

This proxy supports a direct forward calculation from simulated propagation to external magnetic measurements. Conductivity perturbations influence V_m through the simulator; the projection does not multiply its gradient by a spatial conductivity tensor. The source and resulting fields therefore use relative units.

LAT is the openCARP activation threshold crossing at -40 mV. The retained APD80 and APD90 targets measure time from the voltage peak to the first saved sample reaching 80% or 90% repolarization, relative to a resting level averaged over the first five samples. These quantities are in milliseconds and use the final 300 ms beat. APD90 is the primary repolarization endpoint; APD80 provides a consistency check. Fields use linear interpolation with nearest-neighbor fill on a 32×32 tissue grid; simulator perturbations supply the region target.

3.2 Magnetic observation model

For sensor location $\mathbf{r}$ and source element at $\mathbf{r}'$, the direct magnetic calculation follows the Biot–Savart structure

$$\mathbf{B}(\mathbf{r}, t) \propto \sum_e \frac{\mathbf{J}_{\mathrm{eq},e}(t) \times (\mathbf{r} - \mathbf{r}'_e)}{\|\mathbf{r} - \mathbf{r}'_e\|^3} \, \Delta A_e. \tag{2}$$

Here ΔA_e is triangle area. Only in-plane source components contribute to the evaluated B_z. This sheet calculation produces relative-unit signals and approximates magnetic sensing without modeling a particular SQUID or optically pumped magnetometer array.

The nominal sensor plane is 50 mm above the tissue. A 32×32 grid spans a 12 by 12 mm aperture, including a 1 mm margin around the sheet. The four measurement slots hold B_z, an axial gradient, noisy B_z, and noisy masked B_z. The primary variant activates only the first two slots. Its gradient is the difference between B_z at 65 and 35 mm divided by the 30 mm baseline. Each sequence contains 301 time points. Masks select valid values for normalization and zeroing; they are not concatenated as model inputs.

Three measurement rows per simulation produce 1,500 examples. The augmented variants additionally activate the noisy slot, or both noisy and masked slots, while retaining the clean channels. All rows from one simulation stay together. The grouped partition contains 350/75/75 training/validation/test cases, or 1,050/225/225 stored rows. Confirmation uses one predeclared clean row per test simulation: 12 homogeneous, 18 repolarization-only, 18 conduction-only, and 27 mixed cases.

4 Inverse operator models

4.1 Fourier neural operator

The FNO stacks time and channels into 1,204 features and appends three coordinates. A lifting layer maps the input to width 64. Four spectral blocks retain 12 Fourier modes along each spatial axis, combining spectral convolution and a learned pointwise path with nonlinear activation. Output heads

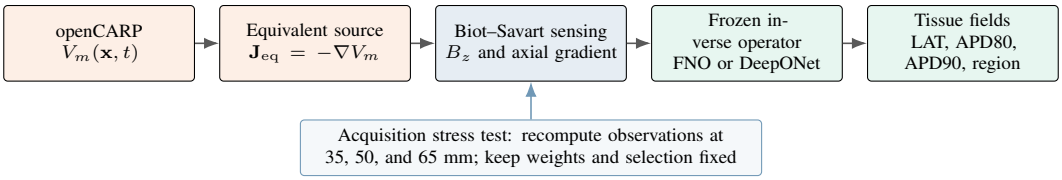


Figure 1: Benchmark pipeline. Cardiac simulation supplies the hidden physical field, a direct forward calculation supplies sensor movies, and a learned inverse operator reconstructs tissue-scale electrophysiology. The distance experiment changes only the synthetic acquisition before frozen-model inference.

predict LAT, APD80, APD90, and region probability on the 32×32 tissue grid. The model has 4.82 million parameter entries.[1]

FNO provides global receptive fields with modest depth. Sensor and tissue grids share a 32×32 resolution, although the sensor aperture is wider. Coordinates encode normalized sensor x, y and height $h/50$ mm. Height is 1.0 throughout training and becomes 0.7 or 1.3 at the alternative test planes, taking this feature outside its training range.

4.2 DeepONet

DeepONet separates the sensor movie from output coordinates. Its branch reshapes $[4, 301, 32, 32]$ to 1,204 spatial channels and uses convolutional widths 48, 64, and 96, beginning with a 1×1 lifting convolution. The trunk uses normalized tissue coordinates with four bands of Fourier features and two hidden layers of width 96. It emits 64 shared basis functions; the branch emits 64 coefficients and a bias per output. Their inner product, divided by $\sqrt{64}$, plus the bias predicts LAT, APD80, APD90, or a region logit. Auxiliary heads and objectives also supervise global region quantities (Appendix B). The model contains 0.287 million parameters.

Training samples 256 of the 1,024 output coordinates per example; evaluation queries the full grid. The branch receives masked, normalized magnetic values without mask channels, sensor coordinates, or height metadata. Its distance response therefore reflects changed measurements after normalization.

4.3 Training and validation locking

Training uses AdamW, learning rate 8×10^{-4}, weight decay 10^{-4}, dropout 0.08, batch size 6, seed 20260525, a ten-epoch cap, and patience three. Both models optimize smooth-L1 field loss and BCE plus soft-Dice region loss; DeepONet additionally uses auxiliary objectives (Appendix B). FNO stops after eight epochs; DeepONet reaches the cap while validation loss and sampled Dice still improve. Inputs use $(B - \mu_s)/\sigma_s$, with one mean and standard deviation over all valid channels, pixels, and frames in each example. Invalid entries become zero. Target scaling uses training statistics.

Selection has two validation-only stages. Within runs, FNO minimizes validation loss and DeepONet minimizes validation loss plus $0.2(1 - \text{Dice})$ on sampled queries. Across four configurations per family, the field representative maximizes mean validation LAT/APD90 R^2. Separate localization representatives maximize validation Dice under the released extraction policy and use measurement-derived descriptors. All headline field, region, and distance results use the same field-selected weights; localization representatives are secondary. A separate refinement model is used only in Section 6.4.

The release retains configurations, checkpoint roles, training logs, grouped splits, and test predictions. We use established models and training methods; the contribution is the benchmark, operator comparison, and locked evaluation protocol.

5 Evaluation protocol

Field reconstruction. Global R^2 is recomputed after pooling all valid target-grid values from the relevant independent simulations. We report LAT and APD90 as primary endpoints and APD80 as a

[1]This count includes inactive auxiliary heads and counts each complex spectral weight once, following PyTorch `numel`. Counting real and imaginary components separately gives 9.54 million real scalar parameters.

Table 1: Performance of field-selected checkpoints at the nominal 50 mm sensor plane. Intervals resample independent test simulations. Dice is pooled over 63 abnormal cases and is included to distinguish global reconstruction from compact-region recovery.

Endpoint	n	FNO	DeepONet
LAT global R^2	75	0.954 [0.941, 0.965]	0.968 [0.960, 0.975]
APD80 global R^2	75	0.957	0.976
APD90 global R^2	75	0.960 [0.944, 0.971]	0.970 [0.966, 0.974]
Abnormal-region Dice	63	0.578 [0.498, 0.648]	0.545 [0.466, 0.614]
Mean per-case Dice	63	0.470	0.452
Zero-overlap cases	63	12	14
Parameter entries		4.82 M	0.287 M

secondary consistency check. A pooled measure weights every valid tissue pixel, which is appropriate for overall reconstruction but can be insensitive to a compact region.

Localized recovery. Region probability is thresholded at 0.5. Pooled Dice aggregates pixels over the 63 abnormal test cases; mean per-case Dice and zero-overlap counts provide secondary diagnostics. APD90 contrast is mean predicted APD90 inside minus outside the predicted region, compared across cases against true APD90 contrast under the true region. Empty or full masks receive zero predicted contrast. Appendix A gives center and false-positive area diagnostics.

Sensor-distance sensitivity. We recompute both magnetic channels for the same 75 test simulations at 35, 50, and 65 mm. The gradient baseline moves with the central plane: its endpoints are 20/50, 35/65, and 50/80 mm. Weights, normalization code, masks, thresholds, and evaluation stay fixed; FNO height coordinates follow the tested plane. Paired changes measure the existing pipeline's acquisition sensitivity. Additional controls compare a training-only mean field and repeat FNO inference with its height coordinate held at 1.0, without retraining or tuning.

Uncertainty. Intervals use 5,000 base-simulation bootstrap samples and two-sided 95% percentiles [28]. Seed 20260901 applies to the primary field and height analyses, 20260903 to Section 6.4, and 20261009 to the training-mean control. Resampling preserves pairing. These intervals describe test-case sampling, excluding retraining and optimization variability.

6 Results

6.1 Accurate reconstruction of activation and repolarization

Both inverse operators reconstruct the two primary fields accurately on unseen simulations. FNO reaches LAT $R^2 = 0.954$ with interval [0.941, 0.965] and APD90 $R^2 = 0.960$ with interval [0.944, 0.971]. DeepONet reaches LAT $R^2 = 0.968$ [0.960, 0.975] and APD90 $R^2 = 0.970$ [0.966, 0.974]. APD80 provides a consistent secondary result, with $R^2 = 0.957$ for FNO and 0.976 for DeepONet.

DeepONet improves LAT R^2 by 0.0139; the paired 95% interval for FNO minus DeepONet is $[-0.0261, -0.0032]$. The APD90 difference favors DeepONet by 0.0105, with interval $[-0.0244, 0.0004]$ for FNO minus DeepONet. Parameter count, coordinates, query strategy, auxiliary objectives, and convergence differ, so this single-seed comparison does not isolate architecture. Both learned models substantially outperform a fixed per-pixel mean fitted only to the 350 training cases, which scores LAT $R^2 = -0.010$ $[-0.028, 0.002]$ and APD90 $R^2 = 0.185$ [0.161, 0.207].

Figure 2 summarizes the main distinction. Global reconstruction remains strong over the full cohort and within the 18 conduction-only cases. Region overlap is much lower, especially for conduction slowing. This result motivates task-level evaluation but does not diminish the central reconstruction outcome: both models recover smooth activation and repolarization fields with high explained variance.

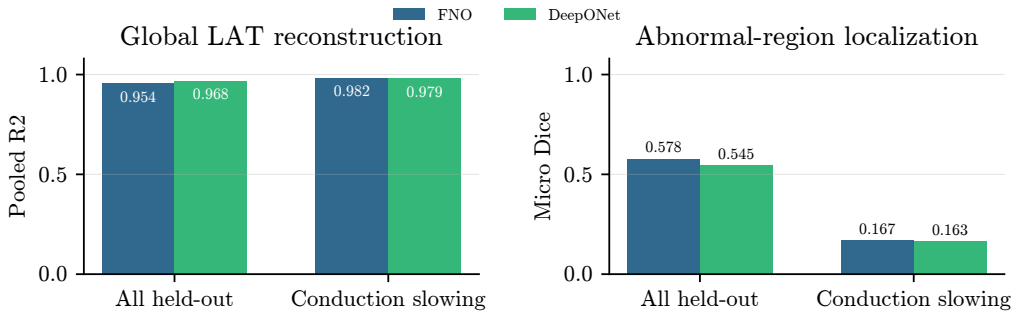


Figure 2: Global LAT reconstruction and abnormal-region localization computed from the same field-selected weights. Full-cohort LAT uses 75 independent test simulations; full-cohort Dice uses 63 abnormal simulations; conduction uses 18 simulations. The figure separates successful field inversion from the harder compact-region task.

Table 2: Frozen-checkpoint field reconstruction across synthetic sensor-to-tissue distance. Values are estimate [95% marginal base-case bootstrap interval]. Height effects use paired differences in the text.

Model	Endpoint	35 mm	50 mm	65 mm
FNO	LAT R^2	0.908 [0.890, 0.922]	0.954 [0.941, 0.965]	0.880 [0.862, 0.895]
FNO	APD90 R^2	0.894 [0.884, 0.903]	0.960 [0.944, 0.971]	0.851 [0.821, 0.877]
DeepONet	LAT R^2	0.967 [0.958, 0.974]	0.968 [0.960, 0.975]	0.968 [0.960, 0.975]
DeepONet	APD90 R^2	0.969 [0.965, 0.973]	0.970 [0.966, 0.974]	0.970 [0.966, 0.974]

6.2 Frozen-checkpoint sensitivity to sensor distance

The frozen pipelines differ in their distance response. DeepONet LAT R^2 equals 0.967, 0.968, and 0.968 at 35, 50, and 65 mm; APD90 R^2 equals 0.969, 0.970, and 0.970. The largest absolute change from 50 mm is 0.00123. At 35 mm, paired LAT/APD90 changes are -0.00123 [-0.00178, -0.00072] and -0.00102 [-0.00149, -0.00052]. These are small but detectable losses. Paired differences, rather than overlap of marginal intervals in Table 2, quantify height effects.

FNO is more sensitive. At 35 mm, LAT/APD90 R^2 fall to 0.908/0.894; at 65 mm they fall to 0.880/0.851. The 65 mm changes are -0.0737 for LAT [-0.0834, -0.0647] and -0.1082 for APD90 [-0.1269, -0.0911]. Among the tested heights, FNO performs best at its training geometry.

Normalization removes a common amplitude scale, while relative channel amplitudes and spatial and temporal patterns still change with height. DeepONet's stability is specific to this pipeline and provides no evidence of learned geometry conditioning. In an additional FNO control, we hold its height coordinate at the nominal value while changing both magnetic channels. LAT/APD90 R^2 remain 0.908/0.893 at 35 mm and 0.881/0.852 at 65 mm, close to the original shifted-coordinate results. Thus coordinate extrapolation alone does not account for the observed degradation. This control separates two input changes for this checkpoint without identifying an intrinsic architectural cause.

Region scores remain much lower than field scores at every height. DeepONet Dice stays near 0.545, while FNO Dice ranges from 0.505 to 0.578. Sensor robustness and downstream task validity therefore require separate endpoints.

6.3 What the reconstructed fields preserve

The 18 conduction-only simulations isolate slowed propagation. The field-selected FNO reaches LAT/APD90 R^2 of 0.982/0.986, and DeepONet reaches 0.979/0.978. The same fixed training-mean maps score -0.007/0.183 on this subset. The operators therefore recover case-dependent field variation beyond this shared template, while pooled R^2 alone remains insufficient for evaluating compact abnormalities.

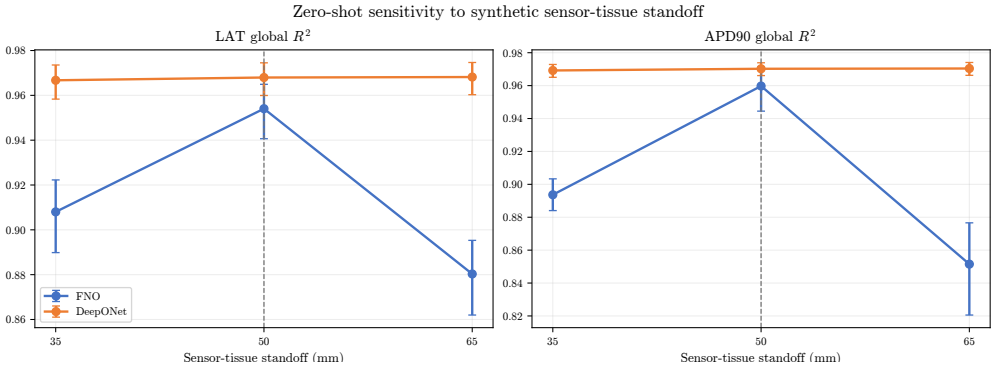


Figure 3: Field sensitivity to synthetic sensor-to-tissue distance without retraining. The dashed line marks the nominal 50 mm plane. Error bars are 95% intervals over the 75 paired simulations.

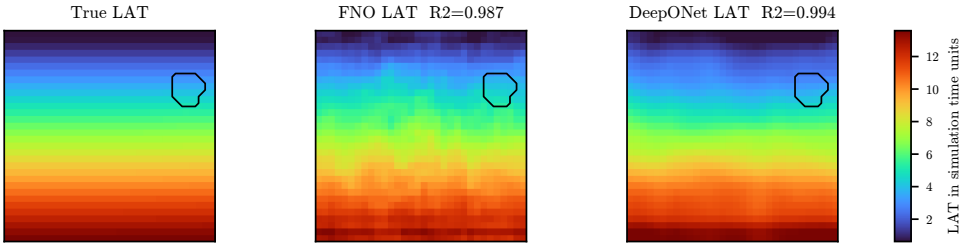


Figure 4: Same-checkpoint LAT example from a conduction-slowing simulation. FNO and DeepONet attain $R^2 = 0.987$ and 0.994. The outlined true intervention supplies spatial context. This disclosed case was selected after analysis and is not a population estimate.

The same checkpoints are less successful at delineating the compact simulator intervention. Their conduction Dice scores are 0.167 for FNO and 0.163 for DeepONet. Each produces an empty or full region in five of the 18 cases. Figure 4 shows the reconstructed LAT maps for a disclosed case selected after analysis. Both operators retain the propagation pattern with R^2 above 0.98.

The population analysis reaches the same qualitative conclusion. Across 63 abnormal cases, pooled Dice is 0.578/0.545 for FNO/DeepONet, and mean per-case Dice is 0.470/0.452, with zero overlap in 12/14 cases. APD90 contrast R^2 is -0.109 [-1.047, 0.337] and 0.095 [-0.572, 0.575]. These wide intervals leave regional contrast recovery uncertain despite accurate global fields.

6.4 Regional diagnostics with known and predicted supports

A separate validation-locked DeepONet refinement model evaluates regional quantities (Appendix B). A 500-tree random forest, the validation-preferred descriptor classifier, uses 50 magnetic descriptors to detect conduction slowing, including mixed cases, with balanced accuracy 0.822 [0.727, 0.907] on 75 test simulations [29]. It uses all 1,050 training rows, minimum leaf size two, balanced-subsample class weights, and seed 20260525. This confirms predictive condition information in the measurements; a global classifier supplies no location.

For the 18 conduction-only cases, we compare the refinement model's predicted LAT contrast with true LAT contrast in milliseconds, each measured inside minus outside the same support. Using the true intervention gives $R^2 = 0.954$ [0.919, 0.974]; using a support derived from true LAT residuals gives 0.948 [0.615, 0.966]. Both supports supply target location and are unavailable in deployment. High contrast agreement conditional on these supports is consistent with useful regional signal, but can also reflect the underlying activation pattern and does not isolate slowing severity.

The refinement model's center-constrained extractor yields 0/18 usable supports. Filling failed contrasts with the predeclared 0 ms value gives $R^2 = -0.0001$ against the true-intervention contrast, a consequence of that fill rule rather than a successful scalar estimate. The contrast between known

and predicted regions motivates separate activation and repolarization support targets. It does not establish a unique causal bottleneck or spatial identifiability.

7 Discussion

7.1 Implications for physical sensor learning

The benchmark illustrates three requirements for learning physical fields from sensors. The independent unit must be the underlying physical case, so every measurement variant from one simulation stays in one partition [27]. Acquisition changes should be tested with frozen checkpoints; retraining at every distance would define a new in-distribution task. Evaluation must also include the quantities that make the reconstruction useful.

Coordinate access is distinct from geometry variation. FNO receives a height coordinate, but it is constant during training. DeepONet queries output coordinates, while its branch receives no sensor-height label. Its stability is empirical behavior under the present normalization and fields. A geometry-conditioned operator should train on varied tissue and sensor layouts, encode acquisition explicitly, and test held-out combinations.

7.2 Why global and local objectives diverge

LAT and APD are smooth over most of the sheet, so a global loss and pooled R^2 are dominated by background tissue. Repolarization appears directly in APD, while conduction slowing appears in LAT timing and gradients. One region head can miss both mechanisms' boundaries. The same issue applies when a globally accurate temperature, pressure, or displacement field misses a source or discontinuity; evaluation must follow the intended decision [22–24].

7.3 Limitations

The simulations use one fixed two-dimensional sheet and one ionic-model family, excluding patient anatomy, torso conduction, three-dimensional geometry, motion, real sensor noise, and measured data. Relative-unit fields use a unit-scaled gradient source. Sensor distance is several times the sheet width, so reconstruction depends on the learned simulator prior rather than demonstrating unique source recovery. The height test varies only planar acquisition, with sample normalization suppressing a common amplitude scale. Clinical validation remains necessary.

Each configuration was trained once, so bootstrap intervals omit optimization variability; height sensitivity also requires replication across seeds. DeepONet was still improving at its ten-epoch cap. The 5,000-case pipeline used eight CPU workers and took 6.13 hours. Training used one RTX 5090 with 32,607 MiB memory, a 16-core AMD Ryzen 9 9950X3D host with 61.6 GiB RAM, PyTorch 2.11.0, and CUDA 12.8. Per-run time and energy were not recorded. The 18-case regional diagnostic is small and uses oracle locations; it cannot establish a universal mechanism.

Broader impact and next experiments. Reliable mapping could support diagnosis, while premature use of synthetic models could misidentify tissue. The release documents failures and limits use to research. No participants, animal data, personal information, or clinical records are used. Next experiments should vary tissue and sensor geometry, train mechanism-specific regions, repeat seeds, and progress to hardware-informed simulation and measured data.

8 Conclusion

This work demonstrates accurate reconstruction of cardiac activation and repolarization fields from simulated magnetic sensor movies. On 75 unseen electrophysiology simulations, FNO reaches LAT/APD90 R^2 of 0.954/0.960 and DeepONet reaches 0.968/0.970. A frozen-checkpoint acquisition test reveals a meaningful distinction: DeepONet remains nearly constant across 35, 50, and 65 mm, while FNO degrades away from the nominal plane. At the same time, compact abnormal-region recovery remains unreliable for conduction slowing. These findings establish a reproducible physical-sensor inversion benchmark and show that field fidelity, acquisition sensitivity, and task-specific localization must be evaluated separately.

References

[1] Gerhard Baule and Richard McFee. Detection of the magnetic field of the heart. *American Heart Journal*, 66(1):95–96, 1963. doi: 10.1016/0002-8703(63)90075-9.

[2] David Cohen. Magnetic fields around the torso: Production by electrical activity of the human heart. *Science*, 156(3775):652–654, 1967. doi: 10.1126/science.156.3775.652.

[3] Riccardo Fenici, Donatella Brisinda, and Anna Maria Meloni. Clinical application of magnetocardiography. *Expert Review of Molecular Diagnostics*, 5(3):291–313, 2005. doi: 10.1586/14737159.5.3.291.

[4] Bradley J. Roth. The magnetocardiogram. *Biophysics Reviews*, 5(2):021305, 2024. doi: 10.1063/5.0201950.

[5] B. N. Cuffin. The role of model and computational experiments in the biomagnetic inverse problem. *Physics in Medicine and Biology*, 32(1):33–42, 1987. doi: 10.1088/0031-9155/32/1/006.

[6] Charulatha Ramanathan, Raja N. Ghanem, Ping Jia, and Yoram Rudy. Noninvasive electrocardiographic imaging for cardiac electrophysiology and arrhythmia. *Nature Medicine*, 10(4):422–428, 2004. doi: 10.1038/nm1011.

[7] Phillip S. Cuculich et al. Noninvasive characterization of epicardial activation in humans with diverse atrial fibrillation patterns. *Circulation*, 122(14):1364–1372, 2010. doi: 10.1161/CIRCULATIONAHA.110.945709.

[8] Yong Wang et al. Noninvasive electroanatomic mapping of human ventricular arrhythmias with electrocardiographic imaging. *Science Translational Medicine*, 3(98):98ra84, 2011. doi: 10.1126/scitranslmed.3002152.

[9] Amel Karoui, Mostafa Bendahmane, and Nejib Zemzemi. Cardiac activation maps reconstruction: A comparative study between data-driven and physics-based methods. *Frontiers in Physiology*, 12:686136, 2021. doi: 10.3389/fphys.2021.686136.

[10] Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023. URL https://jmlr.org/papers/v24/21-1524.html.

[11] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021. URL https://arxiv.org/abs/2010.08895.

[12] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021. doi: 10.1038/s42256-021-00302-5.

[13] R. H. Clayton et al. Models of cardiac tissue electrophysiology: Progress, challenges and open questions. *Progress in Biophysics and Molecular Biology*, 104(1–3):22–48, 2011. doi: 10.1016/j.pbiomolbio.2010.05.008.

[14] Alfonso Bueno-Orovio, Elizabeth M. Cherry, and Flavio H. Fenton. Minimal model for human ventricular action potentials in tissue. *Journal of Theoretical Biology*, 253(3):544–560, 2008. doi: 10.1016/j.jtbi.2008.03.029.

[15] Gernot Plank, Axel Loewe, Aurel Neic, Christoph M. Augustin, Yung-Lin Huang, Matthias A. F. Gsell, Elias Karabelas, Mark Nothstein, Jorge Sánchez, Anton J. Prassl, Gunnar Seemann, and Edward J. Vigmond. The openCARP simulation environment for cardiac electrophysiology. *Computer Methods and Programs in Biomedicine*, 208:106223, 2021. doi: 10.1016/j.cmpb.2021.106223.

[16] Edward J. Vigmond, Rodrigo Weber dos Santos, Anton J. Prassl, Manish Deo, and Gernot Plank. Solvers for the cardiac bidomain equations. *Progress in Biophysics and Molecular Biology*, 96 (1–3):3–18, 2008. doi: 10.1016/j.pbiomolbio.2007.07.012.

[17] Steven A. Niederer et al. Verification of cardiac tissue electrophysiology simulators using an N-version benchmark. *Philosophical Transactions of the Royal Society A*, 369(1954):4331–4351, 2011. doi: 10.1098/rsta.2011.0139.

[18] K. H. W. J. ten Tusscher and A. V. Panfilov. Alternans and spiral breakup in a human ventricular tissue model. *American Journal of Physiology-Heart and Circulatory Physiology*, 291(3): H1088–H1100, 2006. doi: 10.1152/ajpheart.00109.2006.

[19] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.

[20] Sifan Wang, Hanwen Wang, and Paris Perdikaris. Learning the solution operator of parametric partial differential equations with physics-informed DeepONets. *Science Advances*, 7(40): eabi8605, 2021. doi: 10.1126/sciadv.abi8605.

[21] Lee R. Dice. Measures of the amount of ecologic association between species. *Ecology*, 26(3): 297–302, 1945. doi: 10.2307/1932409.

[22] Abdel Aziz Taha and Allan Hanbury. Metrics for evaluating 3d medical image segmentation: Analysis, selection, and tool. *BMC Medical Imaging*, 15:29, 2015. doi: 10.1186/s12880-015-0068-x.

[23] Lena Maier-Hein et al. Why rankings of biomedical image analysis competitions should be interpreted with care. *Nature Communications*, 9:5217, 2018. doi: 10.1038/s41467-018-07619-7.

[24] Annika Reinke et al. Understanding metric-related pitfalls in image analysis validation. *Nature Methods*, 21:182–194, 2024. doi: 10.1038/s41592-023-02150-0.

[25] Romain F. Laine, Ignacio Arganda-Carreras, Ricardo Henriques, and Guillaume Jacquemet. Avoiding a replication crisis in deep-learning-based bioimage analysis. *Nature Methods*, 18: 1136–1144, 2021. doi: 10.1038/s41592-021-01284-3.

[26] Gael Varoquaux and Veronika Cheplygina. Machine learning for medical imaging: Methodological failures and recommendations for the future. *npj Digital Medicine*, 5:48, 2022. doi: 10.1038/s41746-022-00592-y.

[27] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

[28] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7 (1):1–26, 1979. doi: 10.1214/aos/1176344552.

[29] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.

Large language model use disclosure

Large language models were used for manuscript refining, figure generation, and as code assistants. Scripts checked the claims against the saved result files and file integrity records used for the final analysis. The authors remain responsible for the core manuscript writing, scientific claims, analysis, accuracy, and integrity of the submission.

Reproducibility statement

The linked code and result archives contain grouped splits, validation-based selection records, selected checkpoints, training logs, test maps, bootstrap outputs, dependency snapshots, licenses, and checksums. Verification scripts recompute the original primary estimates from saved maps. The accompanying camera-ready supplement, in the `supplemental/` source-package folder, separately provides the additional control script, training IDs and mean maps, coordinate-control predictions, metric tables, and source hashes. Both controls use the same 75 test IDs, and nominal FNO replay agrees within 10^{-5} in R^2. Refitting the mean requires the original 350 training target grids; coordinate inference requires the height-specific measurement tensors. These large inputs and the remaining 4,500 upstream cases are omitted from the compact release. Supplemental maps and `verify_controls.py` support standalone point-estimate verification. There is one training seed per configuration.

Asset licenses

The release documents software licenses. External datasets and openCARP binaries remain outside the release. openCARP is used under Academic Public License 1.1 for noncommercial academic research; carputils uses Apache 2.0 and meshalyzer uses GPLv3. New code is released under MIT and generated result artifacts under CC BY 4.0.

A Evaluation metric definitions

Global R^2 is recomputed within each bootstrap sample after pooling valid pixels. Pooled Dice aggregates pixels across abnormal cases; mean per-case Dice averages overlap separately for each case. Center error is Euclidean distance in normalized sheet coordinates; empty masks receive the maximum grid distance $\sqrt{2}$. At 50 mm, mean center error is 0.211 [0.127, 0.309] for FNO and 0.264 [0.169, 0.375] for DeepONet. Mean false-positive area fraction on 12 homogeneous cases is 0.0258 [0.0075, 0.0465] and 0.0250 [0.0133, 0.0367]. These use the field-selected weights. Separately selected descriptor-augmented localization representatives yield Dice 0.555/0.590 for FNO/DeepONet, reversing the ranking. They are different configurations and are not paired with the headline field scores.

APD90 contrast is the predicted inside-minus-outside mean under the predicted mask, compared by R^2 across 63 abnormal cases with true APD90 contrast under the true intervention. Empty/full predictions receive zero contrast. Section 6.4 instead compares predicted and true LAT inside-minus-outside contrasts under the same diagnostic support, over 18 conduction-only cases. The target LAT-anomaly support uses residuals after fitting a plane to true LAT outside the intervention. Its positive-residual threshold is $\max(0.5P_{75},\, 0.15r_{\max},\, 10^{-6})$, with P_{75} and $r_{\max}$ taken over positive residuals. This support uses target fields and locations.

The training-mean baseline averages each target pixel over the 350 unique training cases, using valid target masks and no measurement or condition information. It is evaluated on exactly the same valid pixels as the neural predictions. Bootstrap samples resample test IDs; the fitted mean stays fixed. This is one shared template, not a stimulus-conditioned reference. The coordinate control reuses the height-specific measurements and masks but assigns each row its original 50 mm coordinate file. At 50 mm, maximum pixel replay differences were 0.0015 ms for LAT and 0.0420 ms for APD90, with absolute R^2 differences below 10^{-5}; numerical replay therefore uses the same metric tolerance as the original height analysis.

B Objectives and diagnostic checkpoints

The field-selected FNO uses smooth-L1 loss with $\beta = 0.75$ on standardized LAT, APD80, and APD90, averaged across fields. Region loss adds binary cross entropy and soft Dice, with twice the BCE weight on homogeneous cases. Field and region weights are both 1.0. Its global parameter, area, and condition heads exist in the parameter count but their auxiliary losses are disabled.

DeepONet uses corresponding field and region losses plus global supervision. Its weights are 1.0 for fields, 1.0 for the region map, 0.45 for presence, 0.70 for region parameters, 0.35 for area category,

0.08 for condition, 0.02 for consistency between area and radius, and 0.04 for scalar loss. Presence uses weighted BCE, parameters use smooth-L1 with $\beta = 0.5$, and area and condition use cross entropy. Consistency penalizes the difference between log area and $\log \pi + 2 \log r$. Scalar loss compares predicted and true APD90 inside-minus-outside means under the true mask at sampled coordinates. These are supervised auxiliary targets, not deployable inputs or physics residuals.

The refinement model is initialized from the field-selected DeepONet and replaces its query outputs with LAT, APD90, region, center heatmap, and distance. Its maximum training budget is eight epochs. Field, region, heatmap, distance, presence, center-parameter, size, and area-bin weights are 0.12, 1.3, 1.4, 0.30, 0.4, 1.2, 0.35, and 0.25. It is selected among four refinement candidates using validation APD90 contrast $R^2 + 0.35\mathrm{Dice} - 0.8$ times homogeneous false-positive area, constrained to Dice ≥ 0.25 and false-positive area ≤ 0.05. Section 6.4 uses the existing center-constrained diagnostic extractor: presence ≥ 0.5, region probability ≥ 0.65, and distance from predicted center ≤ 1.2 times the mean of the radius-head prediction and area-equivalent radius. Nonfinite parameters trigger a plain probability-threshold fallback. This differs from the separately calibrated 0.62 threshold recorded for other regional analyses. It is a disclosed fixed diagnostic, not an optimized deployment method.

C Simulation and measurement settings

The fixed triangular sheet has 10,201 nodes and 20,000 elements, corresponding to a 0.1 mm lattice. Stimuli are applied at the top, bottom, left, or right edge. The dataset manifest records every case's stimulus and intervention parameters. Randomization ranges are patch radius 0.7–2.5 mm, τ_{si} multiplier 1.03–1.45 for repolarization perturbations, conductivity multiplier 0.3–0.9 for conduction perturbations, background longitudinal conductivity parameter 0.85–1.15, and longitudinal-to-transverse conductivity ratio 3–5. The saved final beat spans 0–300 ms in 1 ms output steps. The command records use openCARP integration step 20 microseconds; these are generation settings, not a mesh-convergence validation.

The sensor grid spans 12 mm per axis with spacing $12/31$ mm. Noisy slots add Gaussian noise with standard deviation 5% of clean B_z RMS; the masked slot drops 10% of sensor positions throughout the sequence. The clean B_z and gradient slots remain available in all three variants. Each slot's validity mask is used in the single-scalar normalization. The retained rasterized pilot targets contain no negative failure sentinels for LAT, APD80, or APD90. The APD convention is peak-to-repolarization duration and differs from an activation-to-repolarization interval.

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